Overview of materials for Polyphthalamide (PPA), 30% Glass Fiber Reinforced

Categories: Polymer; Thermoplastic; Polyphthalamide (PPA); Polyphthalamide (PPA), 30% Glass Fiber Reinforced

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polyphthalamide, 30% Glass Fiber Reinforced". Specific grades with glass content between 25% and 34% are included. Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

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Physical Properties	Metric	English	Comments
Density	1.34 - 1.75 g/cc	0.0484 - 0.0632 lb/in³	Average value: 1.49 g/cc Grade Count:121
Filler Content	25.0 - 33.0 %	25.0 - 33.0 %	Average value: 30.5 % Grade Count:32
Water Absorption	0.0200 - 5.00 %	0.0200 - 5.00 %	Average value: 0.369 % Grade Count:49
Moisture Absorption at Equilibrium	0.100 - 2.30 %	0.100 - 2.30 %	Average value: 1.39 % Grade Count:36
	1.25 - 1.35 % @Temperature 70.0 - 70.0 °C	1.25 - 1.35 % @Temperature 158 - 158 °F	Average value: 1.28 % Grade Count:6
Water Absorption at Saturation	4.20 - 4.50 %	4.20 - 4.50 %	Average value: 4.35 % Grade Count:4
Additive Loading	30.0 - 33.0 %	30.0 - 33.0 %	Average value: 30.6 % Grade Count:5
Linear Mold Shrinkage	0.0000200 - 0.0100 cm/cm	0.0000200 - 0.0100 in/in	Average value: 0.00353 cm/cm Grade Count:94
Linear Mold Shrinkage, Transverse	0.000100 - 0.0130 cm/cm	0.000100 - 0.0130 in/in	Average value: 0.00892 cm/cm Grade Count:65
Melt Flow	5.00 - 25.0 g/10 min	5.00 - 25.0 g/10 min	Average value: 12.3 g/10 min Grade Count:3

Mechanical Properties	Metric	English	Comments
Hardness, Rockwell R	114 - 125	114 - 125	Average value: 124 Grade Count:15
Ball Indentation Hardness	230 - 290 MPa	33400 - 42100 psi	Average value: 264 MPa Grade Count:4
Tensile Strength, Ultimate	68.9 - 233 MPa	10000 - 33800 psi	Average value: 173 MPa Grade Cou[illegible]3

	37.0 - 240 MPa @Temperature -40.0 - 200 °C	5370 - 34800 psi @Temperature -40.0 - 392 °F	Average value: 119 MPa Grade Count:25
Tensile Strength, Yield	145 - 225 MPa	21000 - 32600 psi	Average value: 187 MPa Grade Count:12
	63.4 - 140 MPa @Temperature 90.0 - 175 °C	9200 - 20300 psi @Temperature 194 - 347 °F	Average value: 98.4 MPa Grade Count:3
Elongation at Break	1.00 - 4.00 %	1.00 - 4.00 %	Average value: 2.16 % Grade Count:119
	1.50 - 9.00 % @Temperature -40.0 - 200 °C	1.50 - 9.00 % @Temperature -40.0 - 392 °F	Average value: 4.17 % Grade Count:25
Modulus of Elasticity	7.58 - 18.6 GPa	1100 - 2700 ksi	Average value: 11.7 GPa Grade Count:109
	0.400 - 12.0 GPa @Temperature -40.0 - 200 °C	58.0 - 1740 ksi @Temperature -40.0 - 392 °F	Average value: 7.56 GPa Grade Count:28
Flexural Yield Strength	103 - 326 MPa	15000 - 47300 psi	Average value: 255 MPa Grade Count:96
	80.0 - 227 MPa @Temperature 100 - 175 °C	11600 - 32900 psi @Temperature 212 - 347 °F	Average value: 131 MPa Grade Count:7
Flexural Modulus	1.50 - 14.1 GPa	218 - 2050 ksi	Average value: 10.8 GPa Grade Count:97
	3.10 - 9.90 GPa @Temperature 100 - 200 °C	450 - 1440 ksi @Temperature 212 - 392 °F	Average value: 5.66 GPa Grade Count:19
Flexural Strain at Break	2.50 - 3.00 %	2.50 - 3.00 %	Average value: 2.88 % Grade Count:4
Compressive Yield Strength	145 - 276 MPa	21000 - 40000 psi	Average value: 183 MPa Grade Count:9
Poissons Ratio	0.340 - 0.410	0.340 - 0.410	Average value: 0.385 Grade Count:12
Fatigue Strength	100 - 100 MPa @# of Cycles 562 - 562	14500 - 14500 psi @# of Cycles 562 - 562	Average value: 100 MPa Grade Count:1
	100 - 100 MPa @Temperature 140 - 140 °C	14500 - 14500 psi @Temperature 284 - 284 °F	Average value: 100 MPa Grade Count:1
Shear Modulus	4.15 - 5.18 GPa	602 - 751 ksi	Average value: 4.58 GPa Grade Count:3
Shear Strength	70.0 - 101 MPa	10200 - 14600 psi	Average value: 89.5 MPa Grade Count:10
Izod Impact, Notched	0.370 - 3.74 J/cm	0.693 - 7.00 ft-lb/in	Average value: 0.840 J/cm Grade Count:56
Izod Impact, Unnotched	1.60 - 14.0 J/cm	3.00 - 26.2 ft-lb/in	Average value: 6.64 J/cm Grade Count:37
Izod Impact, Notched (ISO)	2.80 - 54.6 kJ/m²	1.33 - 26.0 ft-lb/in²	Average value: 10.3 kJ/m² Grade Count:26
	7.00 - 10.0 kJ/m² @Temperature -30.0 - -30.0 °C	3.33 - 4.76 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 8.50 kJ/m² Grade Count:2
Izod Impact, Unnotched (ISO)	29.0 - 68.0 kJ/m²	13.8 - 32.4 ft-lb/in²	Average value: 46.9 kJ/m² Grade Count:14
	37.0 - 37.0 kJ/m² @Temperature -30.0 - -30.0 °C	17.6 - 17.6 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 37.0 kJ/m² Grade Count:1
Charpy Impact Unnotched	2.60 - 9.70 J/cm²	12.4 - 46.2 ft-lb/in²	Average value: 5.81 J/cm² Grade Count:61
	3.50 - 7.50 J/cm² @Temperature -30.0 - -30.0 °C	16.7 - 35.7 ft-lb/in² @Temperature -22.0 - -22.0 °F	Average value: 4.87 J/cm² Grade Count:15
Charpy Impact, Notched	0.300 - 2.00 J/cm²	1.43 - 9.52 ft-lb/in²	Average value: 0.943 J/cm² Grade Count:63
	0.600 - 2.10 J/cm² @Temperature -40.0 - -30.0 °C	2.86 - 9.99 ft-lb/in² @Temperature -40.0 - -22.0 °F	Average value: 0.935 J/cm² Grade Count:15
Coefficient of Friction	0.220 - 0.290	0.220 - 0.290	Average value: 0.243 Grade Count:3

Coefficient of Friction, Static	0.180 - 0.330	0.180 - 0.330	Average value: 0.230 Grade Count:3
K (wear) Factor	15.0 - 46.3 x 10^{-8} mm³/N-M	7.45 - 23.0 x 10^{-10} in³-min/ft-lb-hr	Average value: 25.4 x 10^{-8} mm³/N-M Grade Count:3

Electrical Properties	Metric	English	Comments
Electrical Resistivity	1000 - 1.00e+17 ohm-cm	1000 - 1.00e+17 ohm-cm	Average value: 4.17e+15 ohm-cm Grade Count:61
Surface Resistance	1.00e+6 - 3.40e+16 ohm	1.00e+6 - 3.40e+16 ohm	Average value: 4.84e+15 ohm Grade Count:24
Dielectric Constant	3.53 - 5.60	3.53 - 5.60	Average value: 4.16 Grade Count:28
Dielectric Strength	18.0 - 45.0 kV/mm	457 - 1140 kV/in	Average value: 28.6 kV/mm Grade Count:41
Dissipation Factor	0.00400 - 0.0220	0.00400 - 0.0220	Average value: 0.0111 Grade Count:19
Arc Resistance	120 - 300 sec	120 - 300 sec	Average value: 183 sec Grade Count:6
Comparative Tracking Index	375 - 800 V	375 - 800 V	Average value: 593 V Grade Count:50
Hot Wire Ignition, HWI	60.0 - 150 sec	60.0 - 150 sec	Average value: 120 sec Grade Count:4
High Amp Arc Ignition, HAI	74.0 - 120 arcs	74.0 - 120 arcs	Average value: 109 arcs Grade Count:4
High Voltage Arc-Tracking Rate, HVTR	0.000 - 16.0 mm/min	0.000 - 0.630 in/min	Average value: 12.8 mm/min Grade Count:6

Thermal Properties	Metric	English	Comments
CTE, linear	15.0 - 72.0 µm/m-°C	8.33 - 40.0 µin/in-°F	Average value: 29.7 µm/m-°C Grade Count:45
	7.00 - 32.0 µm/m-°C @Temperature -40.0 - 250 °C	3.89 - 17.8 µin/in-°F @Temperature -40.0 - 482 °F	Average value: 17.6 µm/m-°C Grade Count:22
CTE, linear, Transverse to Flow	4.50 - 82.0 µm/m-°C	2.50 - 45.6 µin/in-°F	Average value: 59.0 µm/m-°C Grade Count:25
	33.0 - 150 µm/m-°C @Temperature -40.0 - 250 °C	18.3 - 83.3 µin/in-°F @Temperature -40.0 - 482 °F	Average value: 87.1 µm/m-°C Grade Count:22
Thermal Conductivity	0.230 - 0.300 W/m-K	1.60 - 2.08 BTU-in/hr-ft²-°F	Average value: 0.277 W/m-K Grade Count:3
Melting Point	246 - 340 °C	475 - 644 °F	Average value: 316 °C Grade Count:70
Maximum Service Temperature, Air	120 - 270 °C	248 - 518 °F	Average value: 182 °C Grade Count:19
Deflection Temperature at 0.46 MPa (66 psi)	205 - 320 °C	401 - 608 °F	Average value: 297 °C Grade Count:30
Deflection Temperature at 1.8 MPa (264 psi)	155 - 320 °C	311 - 608 °F	Average value: 282 °C Grade Count:103
Deflection Temperature at 8.0 MPa	165 - 225 °C	329 - 437 °F	Average value: 182 °C Grade Count:5
Vicat Softening Point	160 - 290 °C	320 - 554 °F	Average value: 246 °C Grade Count:5
Glass Transition Temp, Tg	95.0 - 135 °C	203 - 275 °F	Average value: 123 °C Grade Count:6
UL RTI, Electrical	130 - 140 °C	266 - 284 °F	Average value: 136 °C Grade Count:5

Flammability, UL94	HB - 5VA	HB - 5VA	Grade Count:78
	V-0 - V-0 @Temperature 1.50 - 1.50 °C	V-0 - V-0 @Temperature 34.7 - 34.7 °F	Grade Count:1
Flame Spread	25.0 - 80.0 mm/min	0.984 - 3.15 in/min	Average value: 43.3 mm/min Grade Count:3
Glow Wire Test	700 - 960 °C	1290 - 1760 °F	Average value: 866 °C Grade Count:8

Optical Properties	Metric	English	Comments
Transmission, Visible	20.0 - 30.0 %	20.0 - 30.0 %	Average value: 26.7 % Grade Count:3

Processing Properties	Metric	English	Comments
Processing Temperature	296 - 340 °C	565 - 644 °F	Average value: 317 °C Grade Count:10
Nozzle Temperature	320 - 345 °C	608 - 653 °F	Average value: 332 °C Grade Count:9
Melt Temperature	240 - 354 °C	464 - 669 °F	Average value: 326 °C Grade Count:81
Mold Temperature	65.0 - 190 °C	149 - 374 °F	Average value: 134 °C Grade Count:81
Drying Temperature	90.0 - 135 °C	194 - 275 °F	Average value: 114 °C Grade Count:68
Moisture Content	0.0200 - 0.200 %	0.0200 - 0.200 %	Average value: 0.0654 % Grade Count:66
Dew Point	-31.7 - -28.9 °C	-25.0 - -20.0 °F	Average value: -30.7 °C Grade Count:14
Injection Pressure	68.9 - 124 MPa	10000 - 18000 psi	Average value: 93.2 MPa Grade Count:22

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's terms of use regarding this information. Click here to view all the property values for this datasheet as they were originally entered into MatWeb.

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